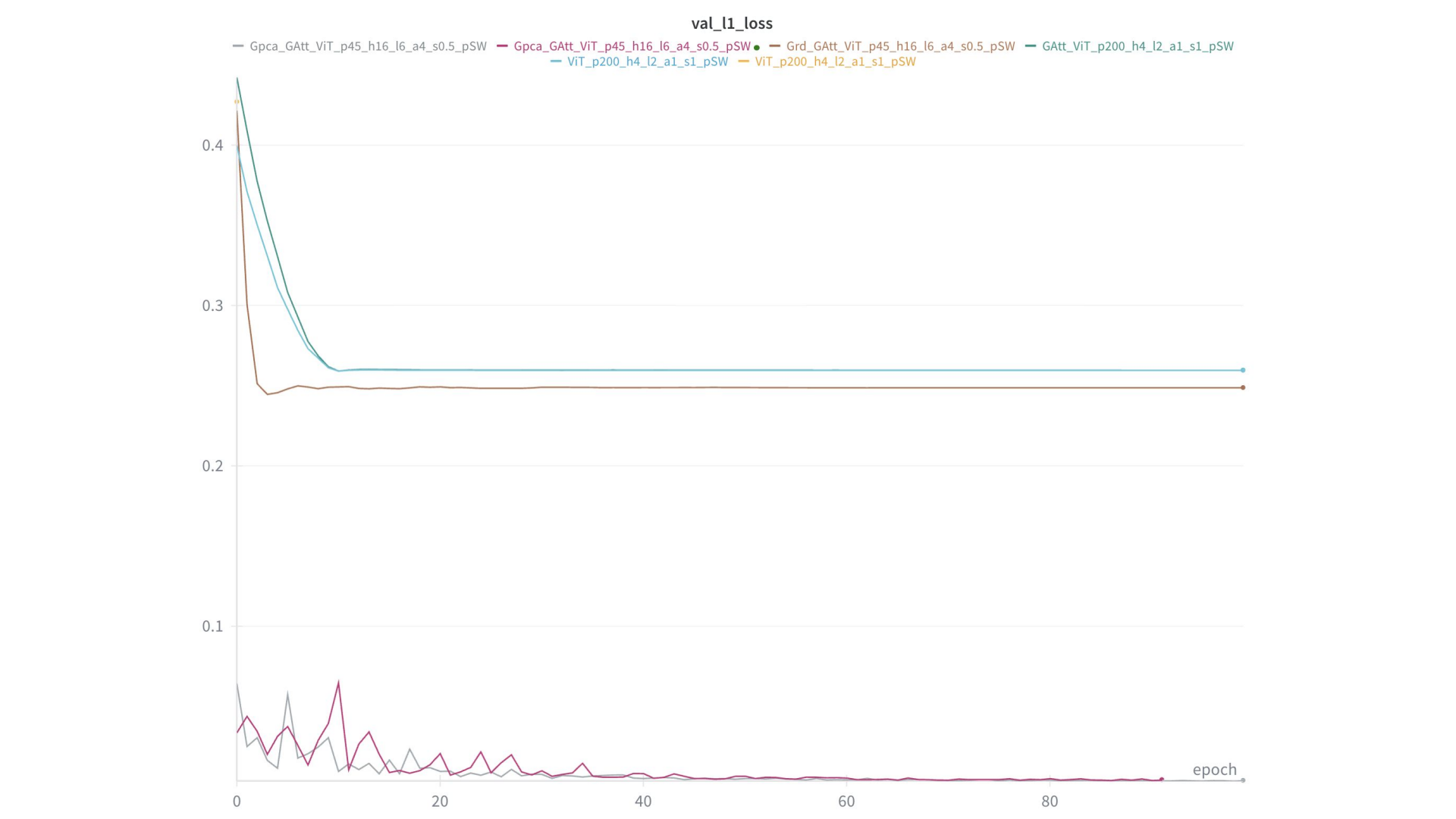
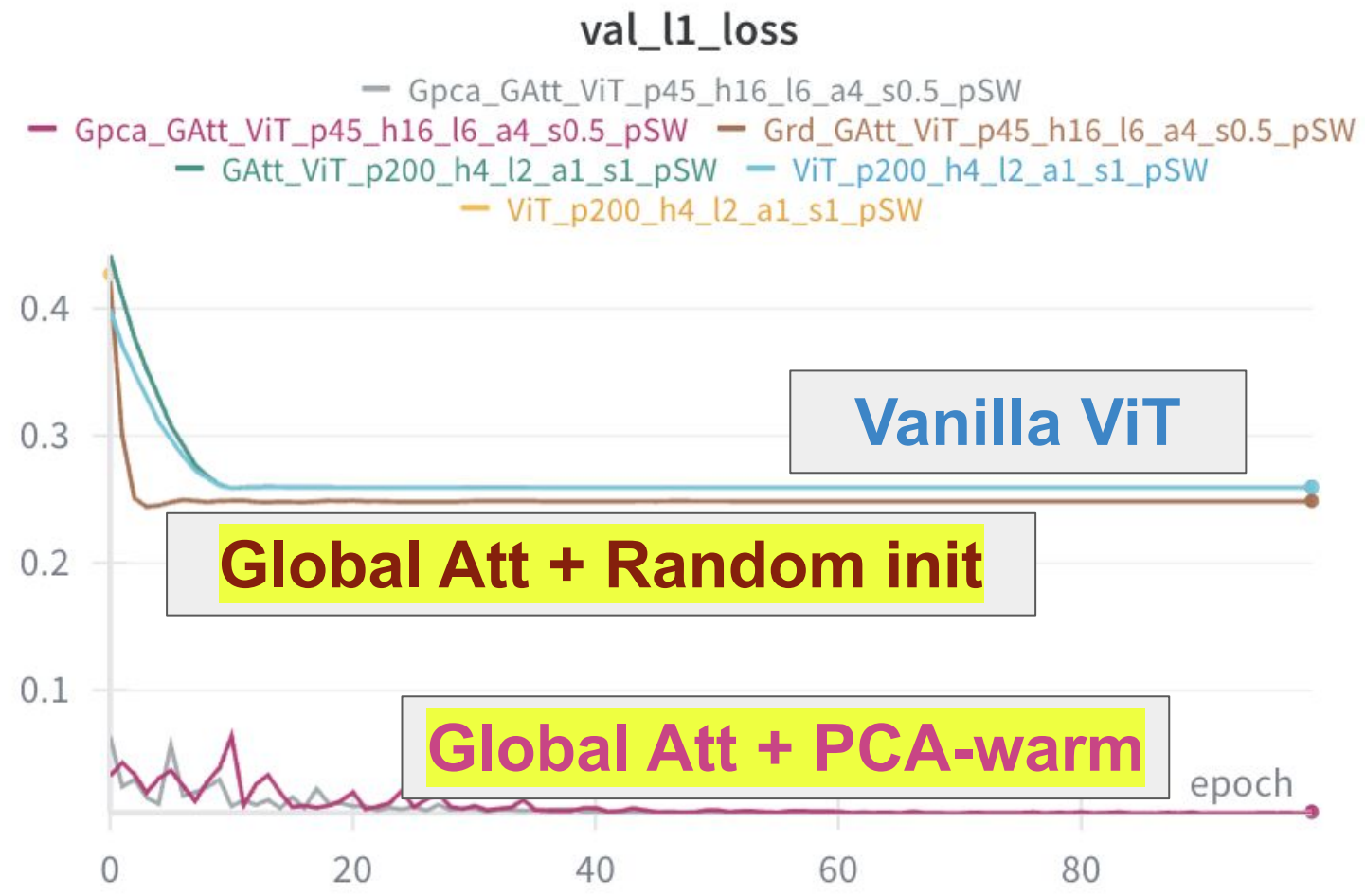
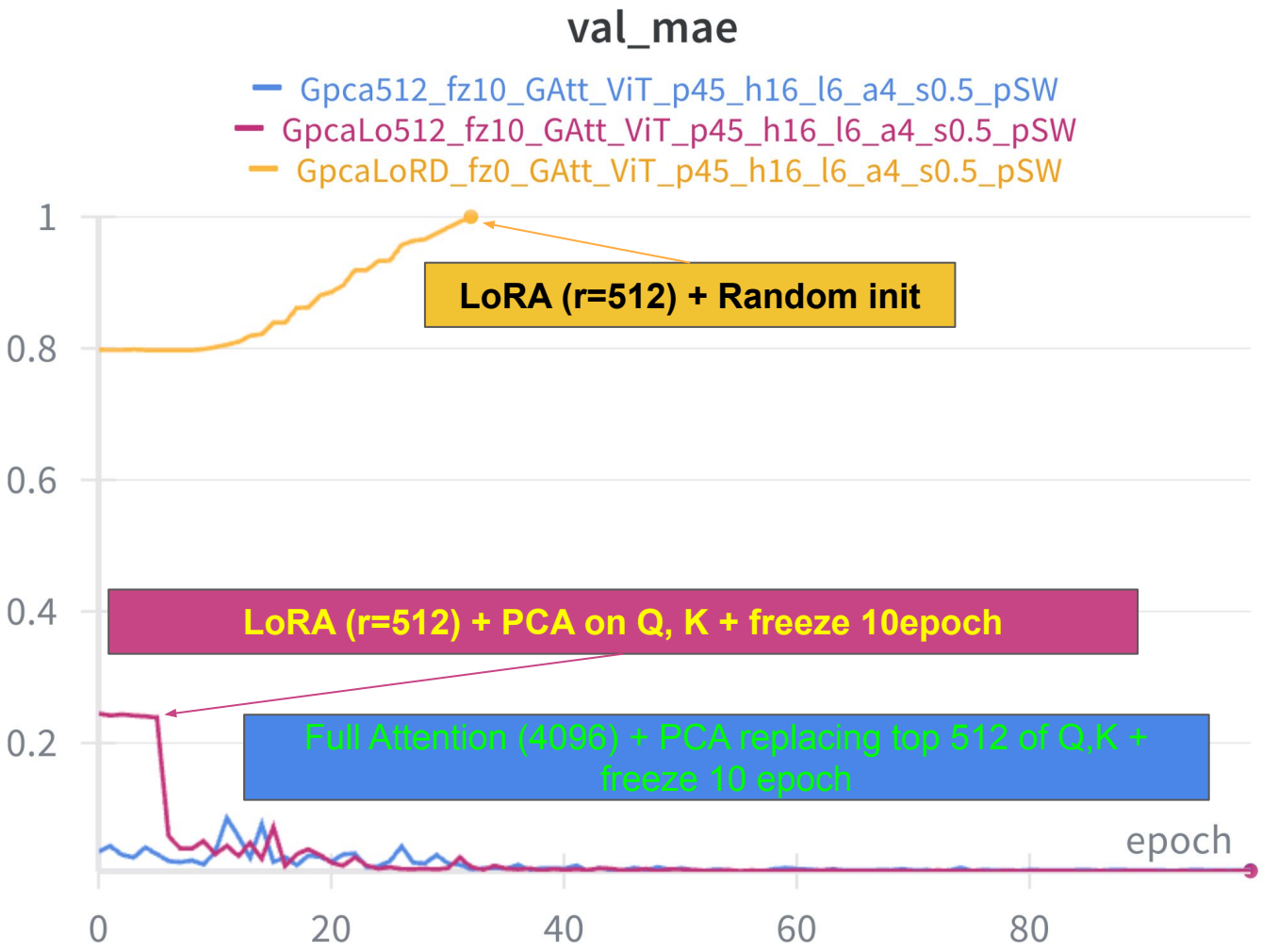

val_l1_loss
Gpca_GAtt_ViT_p45_h16_l6_a4_s0.5_pSW
Gpca_GAtt_ViT_p45_h16_l6_a4_s0.5_pSW
Grd_GAtt_ViT_p45_h16_l6_a4_s0.5_pSW
GAtt_ViT_p200_h4_l2_a1_s1_pSW
ViT_p200_h4_l2_a1_s1_pSW
ViT_p200_h4_l2_a1_s1_pSW
0.4
0.3
0.2
0.1
epoch
0
20
40
60
80


val_l1_loss
Gpca_GAtt_ViT_p45_h16_l6_a4_s0.5_pSW
Gpca_GAtt_ViT_p45_h16_l6_a4_s0.5_pSW
Grd_GAtt_ViT_p45_h16_l6_a4_s0.5_pSW
GAtt_ViT_p200_h4_l2_a1_s1_pSW
ViT_p200_h4_l2_a1_s1_pSW
ViT_p200_h4_l2_a1_s1_pSW
0.4
0.3
0.2
0.1
Vanilla ViT
Global Att + Random init
Global Att + PCA-warm
epoch
0
20
40
60
80


val_mae
Gpca512_fz10_GAtt_ViT_p45_h16_l6_a4_s0.5_pSW
GpcaLo512_fz10_GAtt_ViT_p45_h16_l6_a4_s0.5_pSW
GpcaLoRD_fz0_GAtt_ViT_p45_h16_l6_a4_s0.5_pSW
1
0.8
0.6
0.4
0.2
LoRA (r=512) + Random init
LoRA (r=512) + PCA on Q, K + freeze 10epoch
Full Attention (4096) + PCA replacing top 512 of Q,K + freeze 10 epoch
epoch
0
20
40
60
80

Our experiments show that training with LoRA (r=512) from random initialization is highly unstable and diverges, whereas PCA-based initialization of Q/K enables LoRA to converge within the first 10 epochs using only ~25% of the parameters, with no degradation after unfreezing, achieving performance comparable to full-parameter training. Having validated the effectiveness of **PCA+LoRA**, we can now extend this approach to V/O projections and explore shared vs. per-head configurations; previously infeasible due to ~50M parameters per layer, LoRA (~12M/layer) now makes multi-layer and broader experiments practical.